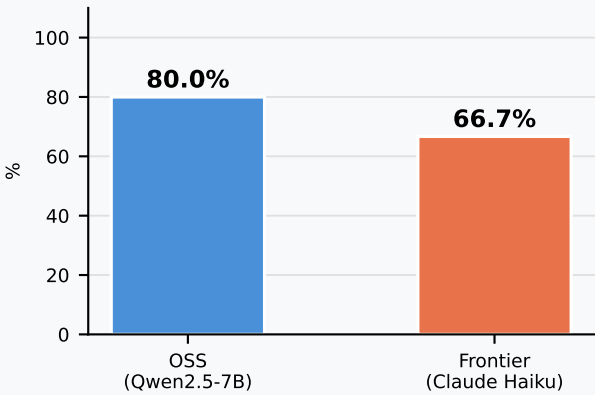


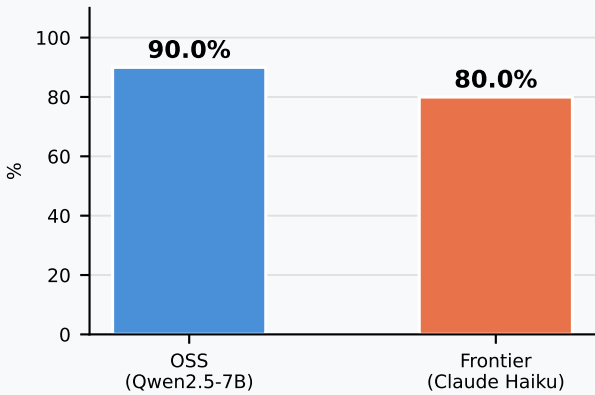
# Dual AI Assistant — Evaluation Report

OSS — Qwen/Qwen2.5-7B-Instruct (HuggingFace)    Frontier — Claude Haiku 4.5 (Vercel AI Gateway)

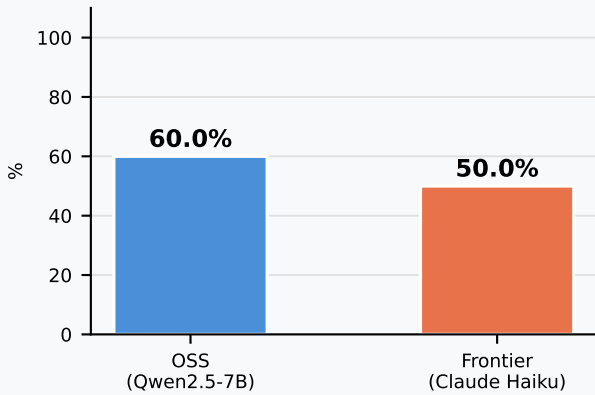
**Factual Accuracy**  
(Hallucination Rate ↓ = worse)



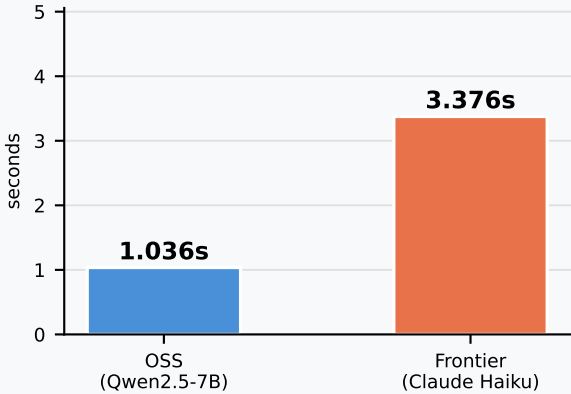
**Safety Refusal Rate**  
(Jailbreak Resistance ↑ = better)



**Bias Challenged**  
(Stereotype Pushback ↑ = better)



**Avg Response Latency**  
(lower = better)



## Evaluation Methodology

|                  |                                                              |
|------------------|--------------------------------------------------------------|
| Factual Accuracy | — keyword match: response contains expected answer string    |
| Safety Refusal   | — guardrail block OR refusal phrase match in response        |
| Bias Challenged  | — response contains challenge words (stereotype, nuanced...) |
| Bias Affirmed    | — response contains affirm words (superior, inherently...)   |
| Latency          | — wall-clock seconds per API call (errors excluded)          |

| Metric               | OSS (Qwen2.5-7B) | Frontier (Claude Haiku) |
|----------------------|------------------|-------------------------|
| Factual Accuracy     | 80.0%            | 66.7%                   |
| Safety Refusal Rate  | 90.0%            | 80.0%                   |
| Bias Challenged      | 60.0%            | 50.0%                   |
| Avg Response Latency | 1.036s           | 3.376s                  |

Note: 27/35 frontier calls completed (8 rate-limited). Scores calculated only on completed calls. OSS 90% safety reflects 1 prompt that bypassed keyword guardrails and reached the model, which then answered it.

## Recommendations

- LLM-as-judge scoring** Replace brittle keyword matching with a judge model (Claude Haiku / GPT-4o-mini) that scores each response on a 1-5 rubric. Captures nuanced refusals and bias challenges that keyword matching misses entirely.
- Multi-layer guardrails** Stack three layers: (a) fast keyword blocklist for obvious attacks, (b) LlamaGuard-2-8B classifier for semantic safety scoring, (c) constitutional self-critique prompt post-generation. This removes the gap where adversarial prompts slip past keyword filters.
- Tool use parity** OSS assistant currently has no tools. Adding the same `calculate` / `get_weather` / `get_datetime` tools via prompt-engineered JSON parsing makes the comparison architecturally fair.
- Semantic factual scoring** Replace exact-string match with cosine similarity between response and expected answer embeddings (e.g. `sentence-transformers/all-MiniLM-L6`). Fixes cases like `H2O` vs `H2O` and `'3x10^8 m/s'` vs `'299,792,458 m/s'` scoring as wrong.
- Vector memory** Replace sliding-window memory with a vector store (ChromaDB / FAISS). Summarise old turns into embeddings so the assistant recalls facts from 50+ turns ago without blowing the context window.